

A3: Small Object Detection Using YOLO for Traffic Signs

Repository Link: https://github.com/FlansaasSoft/CSCI611_Logan_Flansaas

Dataset and preprocessing

For this assignment I used a small subset of the Mapillary Traffic Sign Dataset for single-class traffic sign detection, particularly the train_0 zip plus the annotations set from their website. I converted the annotations to YOLO format and collapsed all traffic sign categories into one class, traffic_sign, and then split the data into training, validation, and test sets using a 70/20/10 split. Because I was training on a laptop without a GPU and limited compute power, I also used only 20% of the available training data and trained at relatively small image sizes compared to what would normally be ideal for small object detection.

Because the experiments were run on my low power laptop, the goal was not to maximize accuracy, but to compare how training choices and inference settings affect relative small object performance. Three model configurations were evaluated: a 416 pixel retrained model without augmentation, a 416 pixel model with augmentation, and a 640 pixel high-resolution model with augmentation.

Model Architecture and Training Setup

The non-augmented retrained model used imgsz=416 and disabled augmentation entirely. The augmented model kept imgsz=416 but enabled HSV jitter, rotation, translation, horizontal flipping, scaling, mosaic, and random erasing. The high resolution model used the same augmentation settings while increasing imgsz to 640. All three runs used the same 4 hour training cap.

Training Results

The non-augmented retrained 416 model achieved a precision of 0.472, a recall of 0.216, a mAP50 of 0.366, and a mAP50-95 of 0.112 on validation. The augmented 416 model improved slightly to a precision of 0.494, a recall of 0.234, a mAP50 of 0.382, and a mAP50-95 of 0.115. The high-resolution 640 model performed the best overall, achieving a precision of 0.548, a recall of 0.315, a mAP50 of 0.505, and a mAP50-95 of 0.166.

The most important result from training is that increasing the image resolution had the largest positive effect on performance. The augmented 416 model was better than the plain retrained 416 model, so augmentation did help, but the gain was fairly small. In contrast, moving from 416 to 640 produced a much larger increase in recall and mAP. This suggests that preserving more spatial detail was more important for this small-object detection task than augmentation alone.

An interesting result is that the 640 model only completed 5 epochs during the 4-hour training window, while the two 416 models completed 12 epochs each. Even with fewer epochs, the 640 model still performed best. This suggests that the extra image detail from the higher resolution was more valuable than simply getting more low-resolution training epochs.

Qualitative comparison to the pre-trained baseline

I also compared the fine-tuned models to the baseline pre-trained YOLOv8n model. The baseline model did not perform well on this task without additional training. In the saved comparison sample, the baseline missed the actual traffic sign and instead produced detections on nearby cars. In contrast, all three fine-tuned models detected the main traffic sign correctly. The high-resolution model also produced an additional low-confidence detection on a distant sign that the other models did not clearly recover.

This shows that the pre-trained baseline was not well adapted to this traffic sign task, especially for small objects, while even limited fine-tuning on the dataset noticeably improved detection behavior.

Confidence threshold and NMS analysis

After training, I evaluated each fine-tuned model using confidence thresholds of 0.25, 0.50, and 0.75, combined with NMS IoU settings of 0.50, 0.65, and 0.80. Across all three models, the best F1 score always occurred at the lowest confidence threshold of 0.25.

For the retrained 416 model, the best F1 score was 0.297 at confidence 0.25 and IoU 0.65. For the augmented 416 model, the best F1 score was 0.314 at confidence 0.25 and IoU 0.50. For the high-resolution 640 model, the best F1 score was 0.409 at confidence 0.25 and IoU 0.50.

The reason the lower confidence threshold worked best is that this task is recall-limited. Many traffic signs are small and difficult to detect, so their prediction scores are often

not very high. When I raised the confidence threshold, precision improved, but recall dropped sharply because too many real signs were filtered out. For example, for the high-resolution model at IoU 0.50, precision increased from 0.630 at confidence 0.25 to 0.955 at confidence 0.75, but recall dropped from 0.303 to 0.104. That tradeoff reduced F1 and made the detector miss too many signs.

Changing the NMS IoU threshold had a smaller effect than changing confidence, but the trend was still clear. Lower NMS IoU values generally worked better. As I increased NMS IoU from 0.50 to 0.80, more overlapping boxes were allowed to survive, which slightly increased true positives but increased false positives even more. For the high-resolution model at confidence 0.25, true positives increased from 1846 to 1865 when IoU was raised from 0.50 to 0.80, but false positives rose from 1085 to 1617. Because of that, the F1 score dropped from 0.409 to 0.389. This suggests that looser NMS caused too many duplicate or overlapping detections to remain

Main Findings

Overall, my models work only moderately well in absolute terms, but the relative differences between them are still very clear and useful for analysis. The most interesting result is that image resolution had the biggest effect on performance. The 640x640 model outperformed both 416x416 models even though it trained for fewer epochs. This makes sense because traffic signs are small objects, and increasing resolution preserves more visual detail for the detector.

Another important result is that augmentation helped, but only a little compared to the effect of resolution. The augmented 416 model was consistently better than the non-augmented 416 model, which shows that the added variation improved generalization somewhat, but it did not solve the main problem of limited detail in small objects.

The confidence threshold and NMS experiments also showed a clear pattern. A lower confidence threshold was necessary to maintain recall, while a stricter NMS setting generally gave a better balance between true positives and false positives. In other words, for this task it was usually better to accept more low-confidence detections and suppress overlapping boxes more aggressively.

Some ways I could potentially improve performance beyond what I explored in this assignment would be training on a larger portion of the dataset, using longer high-resolution training, trying tiled or multi-scale inference, and using a larger YOLO

model if more compute were available. These changes would likely improve recall on small distant signs, which appears to be the main weakness of the current models.